

Set Theory for Computer Science

Week 4: Equivalence Relations

Equivalence relations

Equivalence classes

Equivalence relations v. partitions

Complete system of representatives

Equivalence relations

An **equivalence relation** in A is a relation of type $A \times A$ that is reflexive and transitive and symmetric

Reminder

► reflexivity

$$x R x$$



► transitivity

$$x R y \wedge y R z \rightarrow x R z$$



► symmetry

$$x R y \rightarrow y R x$$



Example: congruence modulo an integer

Congruence modulo m in the integers

$$x \equiv y \pmod{m} \quad \Longleftrightarrow \quad y - x \text{ is a multiple of } m$$

Why an equivalence relation?

- ▶ reflexivity: $x - x = 0 \cdot m$
- ▶ symmetry: $y - x = p \cdot m \quad \Longleftrightarrow \quad x - y = -p \cdot m$
- ▶ transitivity: $y - x = p \cdot m \quad \text{and} \quad z - y = q \cdot m$
 $\Rightarrow z - x = (z - y) + (y - x) = (p + q) \cdot m$

Example: congruence modulo an integer

Congruence modulo m in the integers

$$x \equiv y \pmod{m} \iff y - x \text{ is a multiple of } m$$

Examples

- ▶ Clocks and time: $16 \equiv 4 \pmod{12}$
- ▶ Weekdays: $11 \equiv 25 \pmod{7}$
- ▶ Binary arithmetic: $27 \equiv 283 \pmod{256}$

$$\begin{array}{r} 11001000 \\ 01010011 \\ \hline 00011011 \end{array} + \begin{array}{r} 200 \\ 83 \\ \hline 27 \end{array} +$$

Example: logical equivalence

Logical equivalence of propositional sentences

$\phi \equiv \psi \iff \phi \text{ and } \psi \text{ have the same truth value for every possible valuation}$

Why an equivalence relation?

- ▶ reflexivity: ϕ always has same truth value as ϕ
- ▶ symmetry: immediate from the definition
- ▶ transitivity: if ϕ and ψ have the same truth value and ψ and χ have the same truth value, then ϕ and χ have the same truth value

Example: logical equivalence

Logical equivalence of propositional sentences

$\phi \equiv \psi \iff \phi \text{ and } \psi \text{ have the same truth value for every possible valuation}$

Examples

- ▶ De Morgan:
$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$
$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$
- ▶ distributivity:
$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$
$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$
- ▶ contradictions:
$$\neg p \wedge p \equiv \neg q \wedge q$$

Exercise

In $A := \{a, b, c\}^2 = \{aa, ab, ac, ba, bb, bc, ca, cb, cc\}$, define

$x R y \iff x \text{ and } y \text{ have the same number of } a\text{'s}$

$x Q y \iff x \text{ and } y \text{ start with the same letter}$

- (a) Is R an equivalence relation in A ? And what about Q ?
- (b) List all $x \in A$ such that $x R ab$ holds.
- (c) List all $x \in A$ such that $x Q ab$ holds.

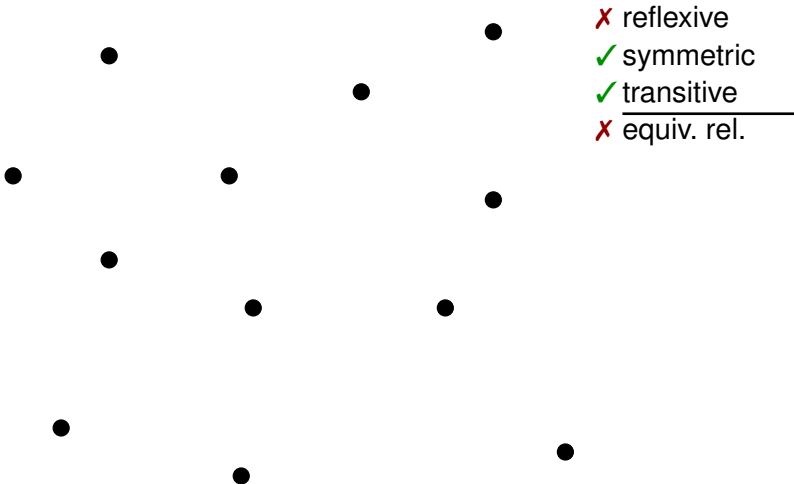
Equivalence relations

Equivalence classes

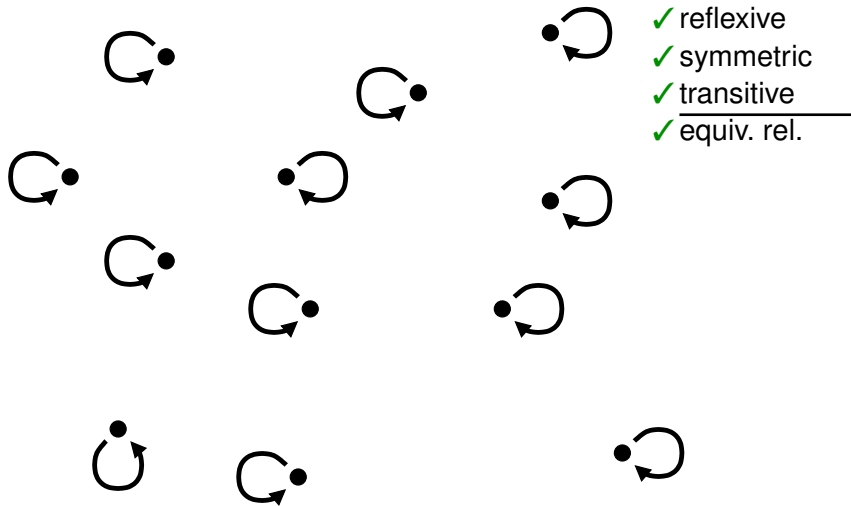
Equivalence relations v. partitions

Complete system of representatives

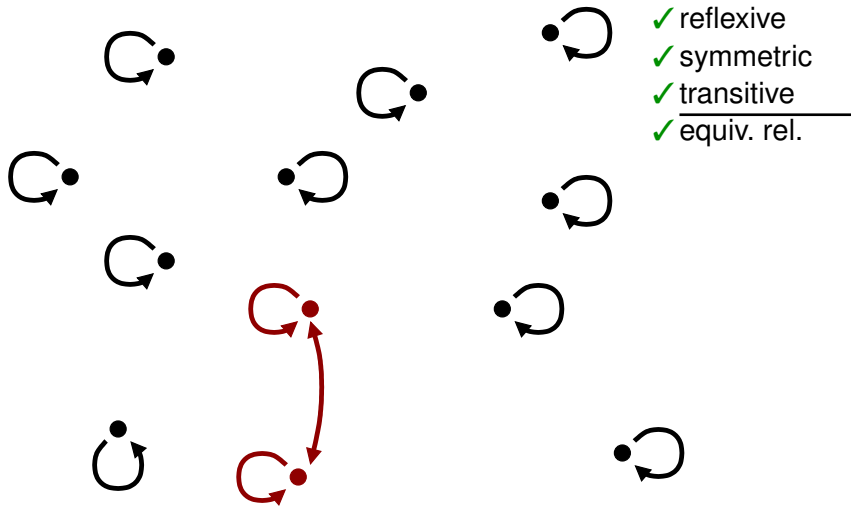
What does an equivalence relation look like?



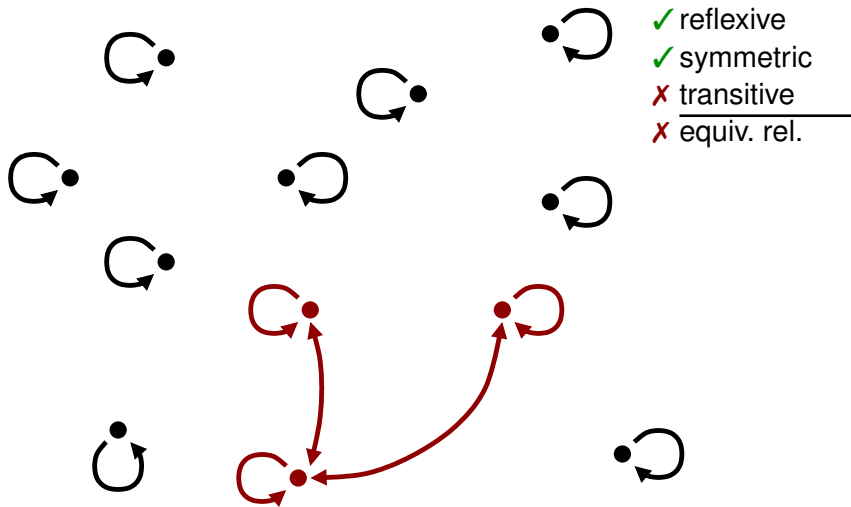
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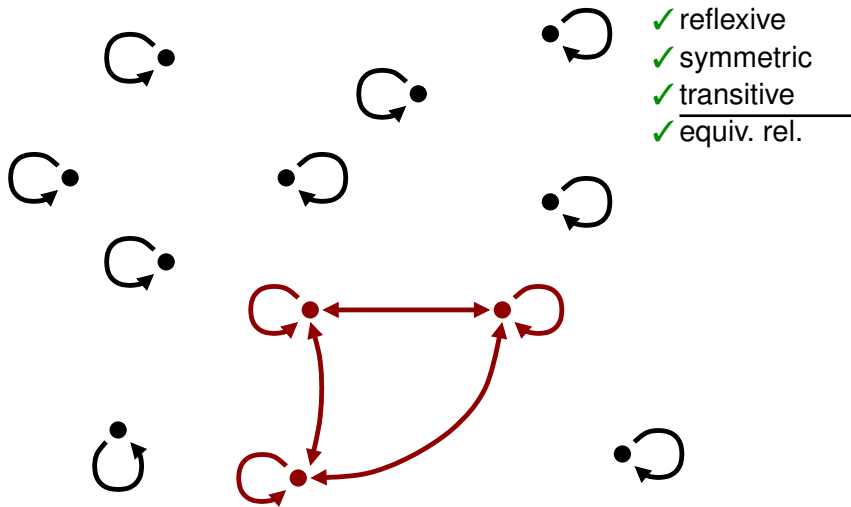
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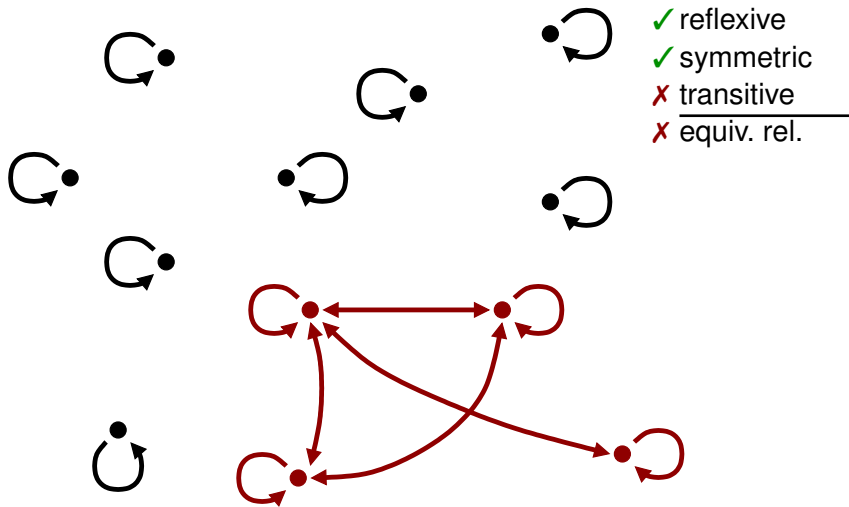
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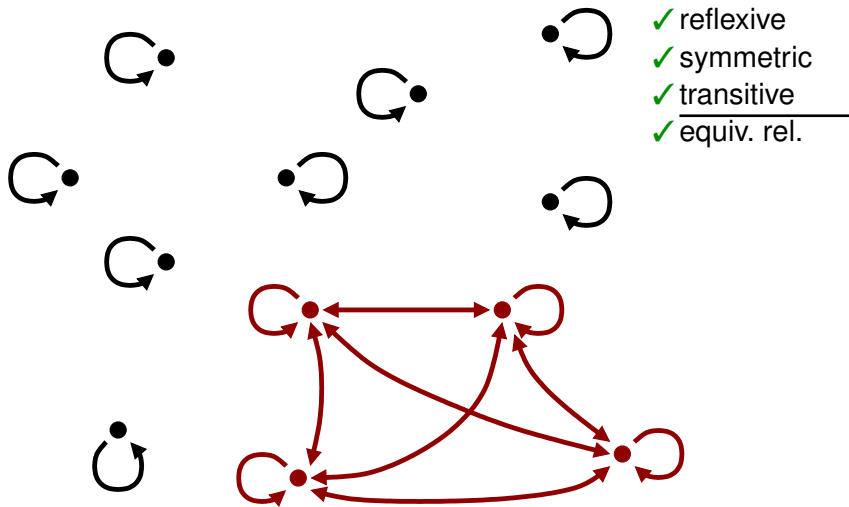
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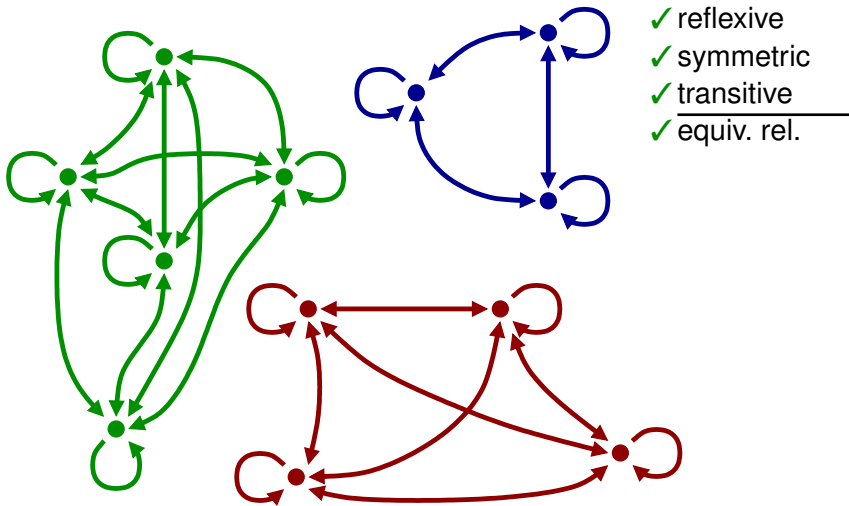
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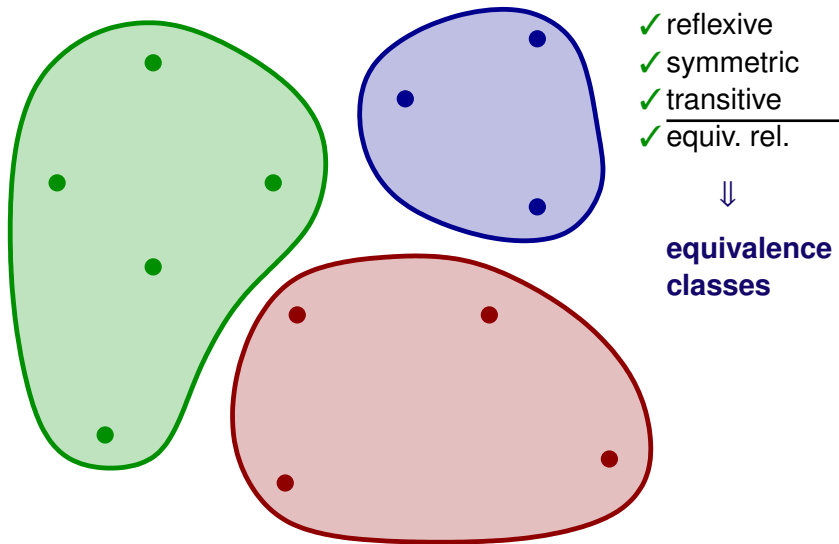
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What does an equivalence relation look like?



Equivalence classes

Let $\equiv$ be an equivalence relation in the set A

The **equivalence class** of an element $a \in A$ is the set

$$[a] := \{x \in A : x \equiv a\}$$

- ▶ all elements of an equivalence class relate to each other
- ▶ elements of different equivalence classes are unrelated

Theorem

The equivalence classes together form a partition of the set A

Example: 3-bit numbers

Example

In the set $A := \{0, 1\}^3$ of 3-bit numbers, define

$$x \equiv y \iff x \text{ and } y \text{ contain the same number of 0's}$$

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Equivalence classes

$[000] = \{000\}$	$[011] = \{011, 101, 110\}$
$[001] = \{001, 010, 100\}$	$[101] = \{011, 101, 110\}$
$[010] = \{001, 010, 100\}$	$[110] = \{011, 101, 110\}$
$[100] = \{001, 010, 100\}$	$[111] = \{111\}$

NB there are *four* equivalence classes (two have three names)

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NB there are *four* equivalence classes (two have three names)

$$\{[a] : a \in A\} = \{[000], [001], [011], [111]\} \text{ is a partition of } A$$

Example: congruence modulo an integer

Example

Consider congruence modulo 3 in $\mathbb{N}$

Example: congruence modulo an integer

Example

Consider congruence modulo 3 in $\mathbb{N}$

Equivalence classes

$$[0] = \{n \in \mathbb{N} : n - 0 \text{ is a multiple of } 3\} = \{3n + 0 : n \in \mathbb{N}\}$$

$$[1] = \{n \in \mathbb{N} : n - 1 \text{ is a multiple of } 3\} = \{3n + 1 : n \in \mathbb{N}\}$$

$$[2] = \{n \in \mathbb{N} : n - 2 \text{ is a multiple of } 3\} = \{3n + 2 : n \in \mathbb{N}\}$$



Example: congruence modulo an integer

Example

Consider congruence modulo 3 in $\mathbb{N}$

Equivalence classes

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$$[2] = \{n \in \mathbb{N} : n - 2 \text{ is a multiple of } 3\} = \{3n + 2 : n \in \mathbb{N}\}$$



$$\{[n] : n \in \mathbb{N}\} = \{[0], [1], [2]\}$$

is a partition of $\mathbb{N}$

Exercise

In $A := \{a, b, c\}^2 = \{aa, ab, ac, ba, bb, bc, ca, cb, cc\}$, define

$x R y \iff x \text{ and } y \text{ have the same number of a's}$

$x Q y \iff x \text{ and } y \text{ start with the same letter}$

- (a) Determine all equivalence classes of the relation R , and write out the partition of A formed by these equivalence classes.
- (b) Do the same for the relation Q .

break

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Complete system of representatives

From equivalence relation to partition

equivalence relation $\equiv$ in A

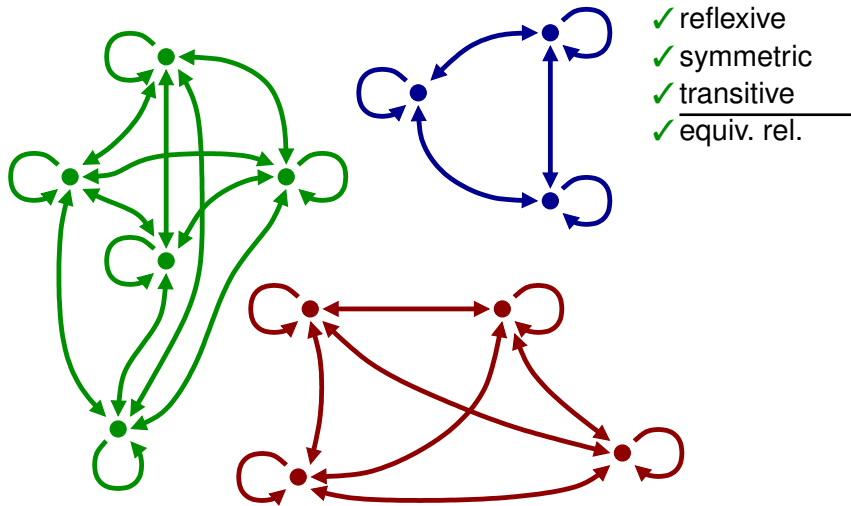


equivalence classes $[a]$

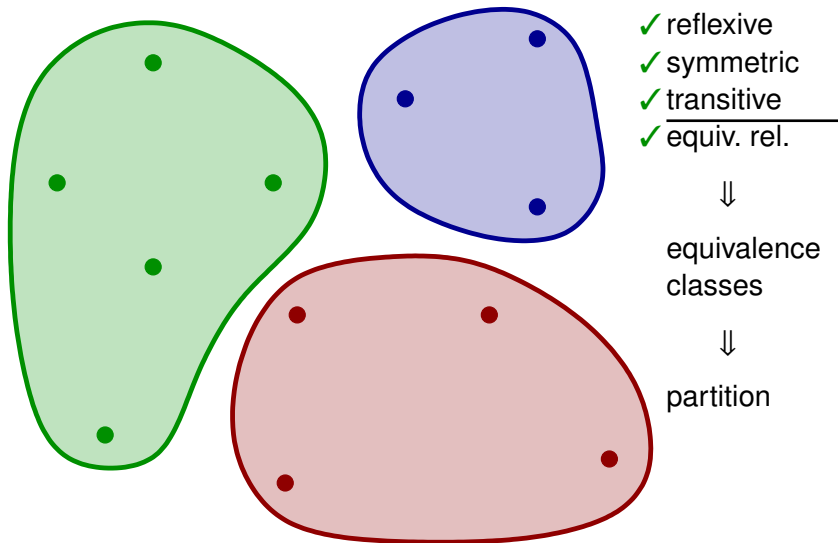


partition $\{[a] : a \in A\}$ of A

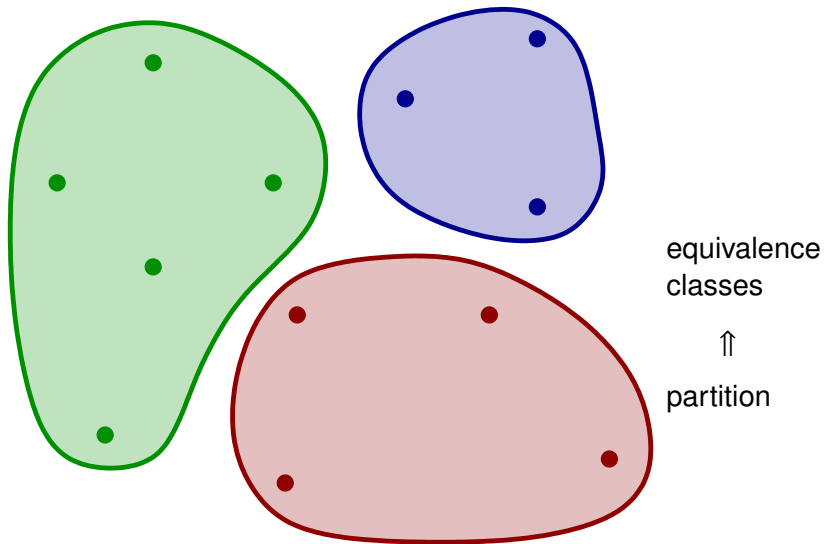
From equivalence relation to partition



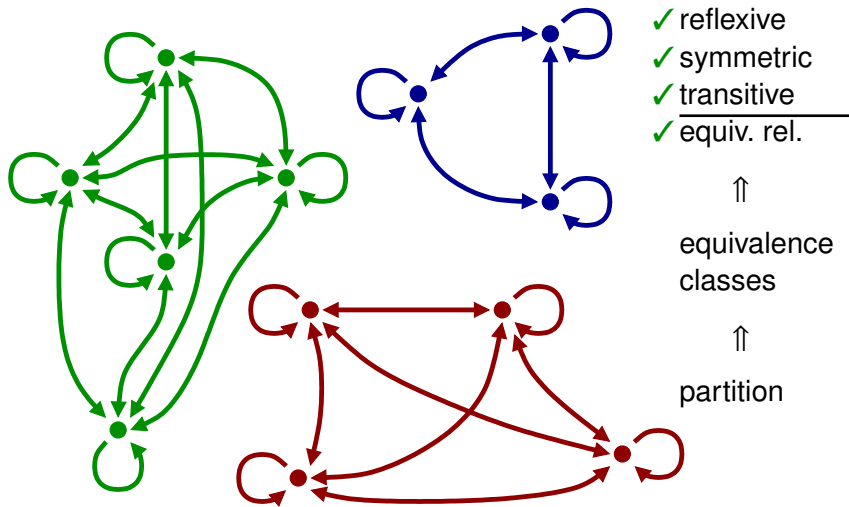
From equivalence relation to partition



From partition to equivalence relation



From partition to equivalence relation



From partition to equivalence relation

partition $P = \{P_1, P_2, \dots\}$ of A



define $\equiv$ in A by

$x \equiv y \iff x \text{ and } y \text{ are in the same part}$



$\equiv$ is an equivalence relation

and $\{[a] : a \in A\} = P$

Example: even and odd integers

Even and odd integers

Partition $\mathbb{N}$ into **even** and **odd** integers

Equivalence relation

The corresponding equivalence relation is congruence modulo 2:

$$x \equiv y \iff x - y \text{ is a multiple of } 2$$



Equivalence classes

$$[0] = \{n \in \mathbb{N} : n \text{ even}\}$$

$$[1] = \{n \in \mathbb{N} : n \text{ odd}\}$$

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Complete system of representatives

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A **complete system of representatives** for $\equiv$ in A is a set $S \subseteq A$ that contains *precisely one* element from each equivalence class

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Example

Consider congruence modulo 3 in $\mathbb{N}$

Partition of equivalence classes

$$\{[n] : n \in \mathbb{N}\} = \{[0], [1], [2]\}$$



Complete system of representatives

$$S = \{0, 1, 2\}$$

(for example)

Complete system of representatives

A **complete system of representatives** for $\equiv$ in A is a set $S \subseteq A$ that contains *precisely one* element from each equivalence class

Example

In the set $A := \{0, 1\}^3$ of 3-bit numbers, consider

$$x \equiv y \quad \Longleftrightarrow \quad x \text{ and } y \text{ have the same number of 0's}$$

Partition of equivalence classes

$$\{[a] : a \in A\} = \{[000], [001], [011], [111]\}$$

Complete system of representatives

$$S = \{000, 001, 011, 111\} \quad (\text{for example})$$

Theorem

A subset S of A is a complete system of representatives for an equivalence relation $\equiv$ in A if and only if

1. every $a \in A$ is equivalent to some $s \in S$
2. two different elements of S are not equivalent

Complete systems of representatives: characterization

Theorem

A subset S of A is a complete system of representatives for an equivalence relation $\equiv$ in A if and only if

1. every $a \in A$ is equivalent to some $s \in S$
2. two different elements of S are not equivalent

Proof ($\Rightarrow$ direction)

1. let $a \in A$ $\Rightarrow$ S contains an element $s \in [a]$
 $\Rightarrow$ $a \equiv s$ for some $s \in S$
2. let $s, s' \in S$ $\Rightarrow$ if $s \neq s'$ then $[s] \neq [s']$
 $\Rightarrow$ if $s \neq s'$ then $s \not\equiv s'$

Theorem

A subset S of A is a complete system of representatives for an equivalence relation $\equiv$ in A if and only if

1. every $a \in A$ is equivalent to some $s \in S$
2. two different elements of S are not equivalent

Proof ($\Leftarrow$ direction)

1. let $a \in A$
 - $\Rightarrow a \equiv s$ for some $s \in S$
 - $\Rightarrow S$ contains an element s from $[a]$
2.
 - $\Rightarrow$ if $s' \in S$ and $s \neq s'$ then $s \not\equiv s'$
 - $\Rightarrow$ if $s' \in S$ and $s \neq s'$ then $s' \notin [s] = [a]$
 - $\Rightarrow S$ contains exactly one element from $[a]$

Exercise

In $A := \{a, b, c\}^2 = \{aa, ab, ac, ba, bb, bc, ca, cb, cc\}$, define

$x R y \iff x \text{ and } y \text{ have the same number of } a\text{'s}$

$x Q y \iff x \text{ and } y \text{ start with the same letter}$

(a) Give a complete system of representatives for R .

(b) Give a complete system of representatives for Q .